先进计算与数字工程研究中心

PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 07 – Foundations of Statistical Learning Theory

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Three periods: evidence, stability, and guarantees

45 minutes	Risk, empirical fitting, train/validation/test splits, and a leakage exercise.
45 minutes	Regularization, bias–variance derivation, and diagnosis from learning curves.
45 minutes	Concentration, finite classes, VC dimension, and structural risk minimization.

One running example

$y = \sin(2\pi x) + \epsilon$, with independent noise $\epsilon \sim \mathcal{N}(0, 0.3^2)$.

Learning outcome

Explain why zero training error can coexist with poor future prediction, and choose an appropriate remedy.

The training winner need not be the future winner

Rule	Training errors / 20	Validation errors / 20	Description
A	0	7	Memorize the 20 records
B	2	3	Use a simple shared pattern

- Training chooses a rule using these records; its error benefits from that choice.
- Validation evaluates records excluded from fitting. Here B is preferable despite its larger training error.
- A single small split is uncertain. We need both a useful model family and an honest evaluation procedure.

Specify the learning problem before fitting

Object	Running regression example
$\mathcal{X}, \mathcal{Y}$	$x \in [0, 1]$; response $y \in \mathbb{R}$
$P(X, Y)$	Unknown joint law producing future input–response pairs
$\mathcal{H}$	Polynomials $f_w(x) = \sum_{j=0}^{m} w_j x^j$
$L(f(x), y)$	Squared error $(f(x) - y)^2$
Learner	A procedure mapping a sample D to fitted coefficients $\hat{w}_D$

Parameters w are fitted. Degree m is selected. An assumption of independent, identically distributed pairs links the observed sample to the target population.

Loss makes different mistakes comparable

Prediction / truth	Absolute loss	Squared loss
$\hat{y} = 8,\ y = 10$	2	4
$\hat{y} = 4,\ y = 10$	6	36

Regression

Squared loss makes a threefold larger residual nine times more costly. The unit becomes the square of the response unit.

Classification

$L(\hat{y}, y) = \mathbf{1}\{\hat{y} \neq y\}$ counts every wrong label equally. A cost matrix can make a missed positive more costly than a false alarm.

Choose loss from the prediction task; an algorithm name does not determine the meaning of an error.

Unequal costs can reverse the preferred class

At one input, the positive-event probability is $p = 0.4$. A missed positive costs four times a false alarm.

$$R(1 \mid x) = 1(1 - 0.4) = 0.6, \qquad R(0 \mid x) = 4(0.4) = 1.6.$$

- Zero–one loss would choose the more probable negative class; the stated cost model instead chooses positive.
- The conditional risk averages the consequences of a proposed decision, not the uncertainty of its parameter estimate.
- First declare what positive means and which mistake is costly. Changing loss can change the optimal action even with the same probabilities.

The best prediction depends on the loss

For a fixed input x, let a be the proposed prediction.

$$\mathbb{E}[(a - Y)^2 \mid x] = (a - \mathbb{E}[Y \mid x])^2 + \mathrm{Var}(Y \mid x).$$

The second term is unaffected by a; squared loss is minimized by the conditional mean.

- Example: $Y = 8$ or 12, each with probability $1/2$. Predicting 10 gives risk 4; predicting 8 gives risk 8.
- Under zero–one loss, predicting class c gives risk $1 - P(Y = c \mid x)$; choose the largest posterior probability.

Squared and absolute loss target different summaries

Suppose $Y = 0$ with probability 0.8 and $Y = 10$ with probability 0.2 at a fixed input.

$$\mathbb{E}[Y] = 2, \quad R_2(2) = 0.8(2^2) + 0.2(8^2) = 16, \quad R_2(0) = 20.$$

- Under absolute loss, predicting zero has risk 2; predicting two has risk 3.2. Zero is a conditional median.
- The conditional mean minimizes squared loss, while a median minimizes absolute loss.
- Neither answer is inconsistent. The loss describes which mistakes we want the predictor to avoid.

Condition on the input before taking the population average

A population has two equally common input groups. The best predictions may differ between them.

$$Y \mid X = 0 \in \{0, 2\}, \quad Y \mid X = 1 \in \{8, 10\} \quad \text{(equal probabilities within each group)}.$$

- Conditional means are 1 and 9; each group has conditional squared risk 1 at its own mean.
- A single constant prediction 5 has risk 17 in either group, although it equals the overall response mean.
- Using an informative input reduces prediction error by separating different conditional distributions. Population risk then averages those conditional risks.

Expected risk and empirical risk answer different questions

$$R(f) = \mathbb{E}_{(X,Y)\sim P}[L(f(X), Y)], \qquad \hat{R}_D(f) = \frac{1}{n}\sum_{i=1}^{n} L(f(x_i), y_i).$$

Quantity	Interpretation
$R(f)$	Average loss over the target population; P is usually unknown
$\hat{R}_D(f)$	Average loss over one observed sample; directly computable

Squared residuals $1, 4, 0, 9$ give empirical risk $14/4 = 3.5$. Another sample could give a different average for the same fixed rule. Large samples reduce sampling fluctuations only when sampling matches the target population.

Two empirical averages need not agree for a fixed rule

A classifier is frozen before both evaluation samples are drawn independently from the same population.

$$\hat{R}_{20} = 3/20 = 0.15, \qquad \hat{R}_{100} = 14/100 = 0.14.$$

- The different observed errors do not imply that the rule changed; sampling alone changes the empirical average.
- A larger representative sample usually reduces estimation uncertainty, not every realized error value.
- Do not choose whichever evaluation sample gives the lower score. Report how sampling was designed and why it represents future inputs.

Empirical risk minimization searches within a chosen family

$$\hat{f}_D = \arg\min_{f \in \mathcal{H}} \hat{R}_D(f).$$

What is optimized

Candidate predictions are compared on observed labels. The hypothesis space and loss have already been specified.

Polynomial fitting

For each degree m, choose its coefficients to minimize average squared residuals. Changing m changes the search space itself.

- ERM does not minimize a directly observed population risk.
- Enlarging a nested family cannot increase its best training fit, but can make the fitted rule less stable.
- A held-out comparison asks whether the extra flexibility predicts unseen observations better.

An interpolating rule wins training without settling prediction

Use observed points $(0,0),(1,1),(2,0)$. Compare a constant rule and a quadratic rule.

$$f_A(x) = 1/3, \quad \mathrm{RSS}_A = 2/3; \qquad f_B(x) = 2x - x^2, \quad \mathrm{RSS}_B = 0.$$

- At $x = 0.5$, the predictions are $1/3$ and 0.75. The three training pairs do not reveal the correct unseen response.
- ERM selects B if both rules are allowed under squared loss; it has minimized the observed objective.
- Useful assumptions or independent observations are needed to decide whether the quadratic behavior between training points is supported.

Three error sources call for three different remedies

Source	Example	Appropriate response
Approximation	A straight line cannot follow a sine wave	Improve features or model family
Estimation	A degree-nine fit changes sharply across small samples	More representative data or stronger regularization
Optimization	Iterations stop far above the attainable training objective	Improve solver, scaling, or stopping criteria

An accurate optimizer can still fit the wrong family. A flexible family can still be estimated poorly. Diagnose the failing step before adding complexity.

Approximation and estimation can be located in one risk diagram

Let f^* be a population-optimal rule and $f^*_{\mathcal{H}}$ the best population rule within the chosen family.

$$R(\hat{f}) - R(f^*) = [R(f^*_{\mathcal{H}}) - R(f^*)] + [R(\hat{f}) - R(f^*_{\mathcal{H}})].$$

- The first difference is the price of restricting the hypothesis family: approximation error.
- The second compares the fitted rule with that family's population oracle; finite data and imperfect computation can contribute.
- Neither oracle is generally observed. We use the decomposition to identify mechanisms, then test remedies through validation and convergence checks.

Memorization exploits the sample, not the shared pattern

Suppose each training record has a unique identifier. A lookup rule returns its stored label and guesses for every new identifier.

- Training zero–one risk is 0 by construction.
- For balanced independent binary labels, guessing on new records has expected error $1/2$.
- A useful inductive bias restricts predictions beyond observed points: smoothness, locality, a linear relation, or a relevant representation.

The missing step

Successful learning requires structure that connects observed and unobserved inputs. A perfect lookup table supplies no such connection.

The same fitted endpoints allow different unseen behavior

Two rules agree on training inputs zero and one but differ between them.

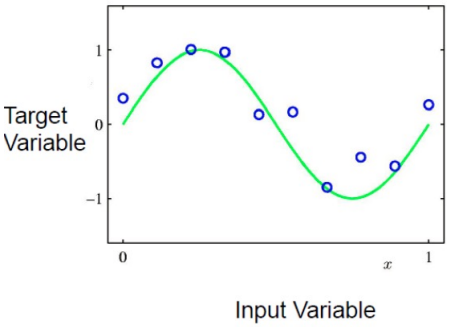
$$f_A(x) = x, \qquad f_B(x) = x + 4x(x-1), \qquad f_A(0.5) = 0.5, \quad f_B(0.5) = -0.5.$$

- Both return 0 at $x = 0$ and 1 at $x = 1$. Training agreement does not specify interpolation behavior.
- A smooth, nearly linear relation is one possible inductive preference; it must fit the actual problem.
- The more complicated rule is not logically excluded by two observations. Additional structure or observations distinguish the predictions.

A controlled example separates signal from noise

$$y = \sin(2\pi x) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 0.3^2).$$

- Ten training inputs are sampled from $[0, 1]$.
- Green signal is the conditional mean; observed points include random noise.
- The learner observes points, not the true curve.
- Judge a fit by independent prediction, not by its ability to hit every point.



Observed responses and the mean signal are different targets

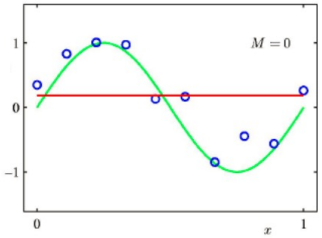
At $x = 0.25$, the running sine signal has mean one; separate noise realizations can give different observations.

$$f^*(0.25) = \sin(\pi/2) = 1, \qquad y_a = 0.7, \quad y_b = 1.2.$$

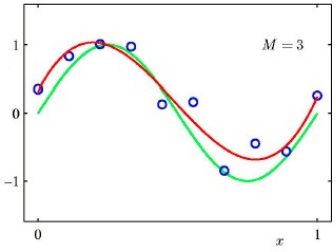
- Predicting the mean leaves residuals -0.3 and 0.2 on these two observations; a nonzero residual is not automatically a modeling failure.
- A rule forced to match 0.7 exactly may move away from the conditional mean.
- The signal is known in this teaching simulation. In a real task, repeated evidence and an appropriate noise model are needed to distinguish signal from fluctuation.

Compare three fits on the same ten observations

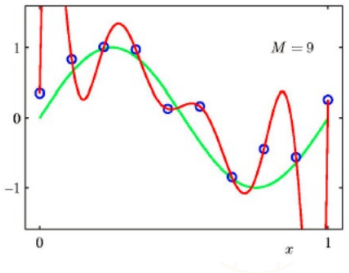
Degree 0



Degree 3



Degree 9



- Degree 0 misses the systematic rise and fall: underfitting.
- Degree 3 follows the main trend without chasing every noise displacement.
- Degree 9 can interpolate ten distinct inputs; its behavior between points can be unstable. Better training fit does not establish better prediction.

Ten coefficients can interpolate ten distinct inputs

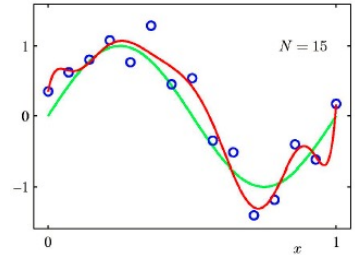
For degree nine, the polynomial design matrix has ten columns: $1, x, \ldots, x^9$.

$$X_{ij} = x_i^{j-1}, \quad i, j = 1, \ldots, 10; \qquad Xw = y.$$

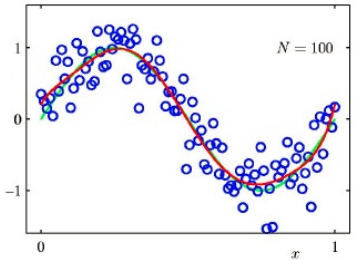
- With ten distinct inputs, this square Vandermonde matrix is nonsingular in exact arithmetic, so an interpolating coefficient vector exists.
- The labels include noise. Exact interpolation therefore fits the noise realizations as well as the signal.
- Numerical conditioning can still be poor. Existence of an exact fit says nothing by itself about stability or error at unseen inputs.

More data changes the usable amount of flexibility

15 observations



100 observations



- Degree nine stays fixed. More observations constrain the curve between isolated points and can reduce sample sensitivity.
- Additional records must represent the intended future population; duplicating a record adds no independent evidence.
- Sample size and capacity interact. No universal samples-per-parameter rule guarantees prediction quality.

Assign separate jobs to training, validation, and testing

Subset	Allowed job
Training	Fit coefficients and learn preprocessing statistics
Validation	Choose degree, penalty strength, or a decision threshold
Test	Estimate performance after those choices are frozen

For 1,000 independent records, an illustrative split is 600/200/200. A temporal task instead trains on earlier records and tests on later ones. Patient, device, or subject groups must remain in a single side of a split when predicting new groups. Stratification preserves class proportions; it does not remove group dependence.

Random and temporal splits can estimate different futures

Suppose a sensor is recalibrated after month six. The intended task is to predict month seven.

- A random split mixing months one through seven can put recalibrated examples into training; it estimates interpolation across the observed mixture.
- Training on months one through six and evaluating month seven asks the stated future-prediction question.
- Different results can reflect a changed population rather than a defective metric. Declare the prediction time and which information is available before fitting.

Cross-validation rotates the held-out observations

For each candidate setting, fit five times. Each fold is predicted by a model fitted on the other four folds.

$$\widehat{R}_{\mathrm{CV}} = \frac{\sum_{k=1}^{5} \sum_{i \in I_k} L(f_{-k}(x_i), y_i)}{\sum_{k=1}^{5} |I_k|}.$$

Equal folds

Fold MSEs $0.10, 0.14, 0.12, 0.09, 0.15$ average to 0.12. Unequal fold sizes require weighting by observation count.

- All folds use the same candidate settings and prediction target.
- After selecting settings, refit on the available development data.
- A separate test set, or nested outer folds, evaluates the entire selection procedure.

Unequal folds require observation-weighted averaging

Three held-out folds contain 10, 20, and 30 observations, with mean losses $0.1, 0.2, 0.3$.

$$\widehat{R}_{\mathrm{CV}} = \frac{10(0.1) + 20(0.2) + 30(0.3)}{60} = \frac{14}{60} \approx 0.233.$$

- An unweighted average of fold means is 0.2; it gives the smallest fold the same influence as the largest.
- For average loss per observation, accumulate the held-out loss sum and total held-out count.
- Choose the aggregation unit deliberately. Equal weighting of patients or groups can answer a different question than equal weighting of visits.

Nested validation evaluates the selection process

An outer fold is held out while inner folds choose a polynomial degree and penalty using only the remaining development data.

- Outer fold 1 may choose degree 3, while outer fold 2 chooses degree 4; the procedure, not one preselected coefficient vector, is being assessed.
- Outer losses $0.12, 0.16, 0.10$ on equal-size folds average to about 0.127.
- After assessment, refit the selected workflow on the available development data. Outer outcomes must not feed back into its inner choices.

Every fitted preprocessing step belongs inside a fold

Tempting procedure	Why the estimate becomes optimistic
Standardize all rows first	Held-out feature statistics enter training
Select features using all labels	Held-out outcomes influence the representation
Duplicate minority rows before splitting	The same evidence can appear on both sides

Correct order

Split first. Fit transformations only on the training part; apply those fitted transformations to its held-out part.

Pipeline

Scaler $\rightarrow$ feature transform $\rightarrow$ estimator is fitted separately within each fold. A test row contributes no fitted statistic.

A fitted scaler can reveal a concrete leakage path

Training feature values are $0, 2$; held-out values are $100, 102$. Compare the means used for centering.

$$\mu_{\text{train}} = 1, \qquad \mu_{\text{all}} = 51.$$

- Training-only centering maps training values to $-1, 1$; all-row centering maps them to $-51, -49$.
- The second transformation has used the held-out feature distribution before model fitting.
- Leakage can occur without directly reading held-out labels. Fit learned transformations on training rows; apply their fixed statistics to held-out rows. A target-distribution shift is then visible instead of hidden.

Group splitting follows the unit that should remain unseen

There are 100 patients with ten visits each. The goal is prediction for a patient never observed during training.

- Randomly separating 1,000 visits can put the same patient's characteristic measurements on both sides.
- Group by patient identifier before assigning train and validation sets; all ten visits from one patient stay together.
- For a different task, predicting a known patient's next visit, a within-patient temporal design may be appropriate. The correct independence unit follows the intended use, not merely the spreadsheet row.

A validation winner still carries selection uncertainty

Suppose 50 candidates have similar true performance. Some will score unusually well on one small validation sample by chance.

- Selecting the smallest observed error favors both genuine improvements and lucky fluctuations.
- Reusing that sample for further feature design makes it part of the development process.
- On an independent test of $n = 200$, an observed error $\hat{p} = 0.10$ has approximate standard error $\sqrt{\hat{p}(1-\hat{p})/n} = 0.021$.

Interpretation

A two-point accuracy difference can be smaller than sampling uncertainty. Prefer stable comparisons and evaluate the frozen procedure independently.

Sampling uncertainty shrinks with the square root of count

For an independently assessed binary error rate of 0.1, use the approximate binomial standard error.

$$\mathrm{SE}(\hat{p}) \approx \sqrt{\hat{p}(1-\hat{p})/n}: \quad n = 200 \Rightarrow 0.0212, \quad n = 800 \Rightarrow 0.0106.$$

- Four times the observations halves this standard error; it does not quarter it.
- The approximation describes a fixed assessment design with independent records and enough observed events.
- A small performance difference can remain ambiguous. More representative evaluation data improves precision; repeated tuning on the same test rows does not supply independent evidence.

Period 1 exercise: repair an evaluation design

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Proposed experiment

There are 300 patients, each with four visits. All 1,200 visits are standardized together, randomly split, and used to compare 80 polynomial settings. The same held-out split is reused after every change.

1. Identify two separate ways that held-out information enters training or selection.
2. Redesign the split for prediction on a new patient.
3. Explain which dataset chooses degree and which estimates final error.

Expected reasoning

Keep patient groups together; fit preprocessing inside training folds; tune on development folds; freeze choices before final testing. A smaller honest score is more informative than a leaked high score.

Regularization trades training fit for a controlled rule

$$J(w) = \underbrace{\frac{1}{n}\sum_i (f_w(x_i) - y_i)^2}_{\text{observed fit}} + \lambda \underbrace{\Omega(w)}_{\text{penalty}}, \quad \lambda \geq 0.$$

- The penalty discourages particular kinds of flexibility; it is not an extra observed prediction error.
- $\Omega(w) = \sum_{j \geq 1} w_j^2$ shrinks slopes and basis coefficients. The intercept is usually excluded.
- A larger λ can accept worse training fit to reduce sensitivity to the sample.
- Choose λ by validation. Extremely strong regularization can erase a real signal.

A penalty can change which candidate has the smaller objective

Compare two fitted candidates using average squared loss plus an L2 penalty.

$$J_A = 0.02 + 100\lambda, \qquad J_B = 0.06 + 4\lambda; \qquad J_A = J_B \Rightarrow \lambda = \frac{0.04}{96}.$$

- A fits better but uses much larger coefficients. Above $\lambda \approx 0.000417$, B has the lower objective.
- At $\lambda = 0.001$, objectives are 0.12 and 0.064 respectively.
- These are training criteria, not held-out prediction risks. Validation determines whether the preference improves future prediction under the chosen feature scaling.

Penalty and constraint express related preferences

Two ways to restrict a fit:

$$\min_w \hat{R}(w) + \lambda\Omega(w), \qquad \min_w \hat{R}(w) \text{ subject to } \Omega(w) \leq t.$$

- The constrained form supplies a budget t; the penalized form supplies a price λ.
- Under appropriate convexity and duality conditions, corresponding solutions can be related through a Lagrange multiplier.
- Smaller budgets or larger prices typically impose stronger restrictions. Their numerical values are not interchangeable.
- A coefficient budget changes when features change units; scale features before interpreting the restriction.

A coefficient budget and a coefficient price can share a solution

Use a one-coefficient squared fitting objective whose unconstrained optimum is three.

$$\min_{|w|\le 1} (w-3)^2 \Rightarrow \hat{w} = 1; \qquad \min_{w} (w-3)^2 + 2w^2 \Rightarrow \hat{w} = 1.$$

- The constrained fit takes the closest feasible point to three; its budget is one.
- The penalized derivative is $2(w-3) + 4w = 0$, yielding the same optimum with price two.
- The budget and price are different numbers. Their correspondence comes from this objective and the active constraint, rather than a universal equality of hyperparameters.

L2 shrinkage can be calculated in one line

Consider one centered coefficient with unpenalized estimate 3:

$$J(w) = (w-3)^2 + \lambda w^2, \quad J'(w) = 2(w-3) + 2\lambda w.$$

Setting the derivative to zero gives $\hat{w} = 3/(1+\lambda)$.

λ	0	1	5
$\hat{w}$	3	1.5	0.5

The fit is smoothly pulled toward zero. This simple calculation describes shrinkage in a decoupled direction; correlated multivariate coefficients can move in more complicated ways. Lower coefficient magnitude can reduce sensitivity to input and sampling noise.

Ridge shrinks weak directions more strongly in relative terms

For two orthogonal feature directions, let their information matrix be $\mathrm{diag}(1, 100)$.

$$Z^\top y = (2, 20)^\top : \quad \hat{w}_{\mathrm{OLS}} = (2, 0.2)^\top, \quad \hat{w}_{\lambda=9} = (0.2, 20/109)^\top.$$

- The direction supported by information one shrinks by a factor $1/10$; information one hundred shrinks by $100/109$.
- Ridge adds the same price to each standardized direction, which has a larger relative effect where data support is weak.
- This decoupled example explains stability. In correlated original coordinates, individual coefficients need not all change monotonically.

L1 can set a coefficient exactly to zero

Use the comparable objective

$$J(w) = \tfrac{1}{2}(w-3)^2 + \lambda|w|.$$

For $w > 0$, stationarity gives $w - 3 + \lambda = 0$. At $w = 0$, the subgradient of $|w|$ is $[-1, 1]$.

$$\hat{w} = \max(3-\lambda, 0): \quad \lambda = 1 \Rightarrow \hat{w} = 2, \qquad \lambda = 4 \Rightarrow \hat{w} = 0.$$

- The kink permits a range of forces to balance at zero: sparse solutions.
- Absolute value is continuous but not differentiable at zero.
- Correlated features can make selection unstable; a zero coefficient is not proof that the variable has no causal effect.

Soft thresholding handles either sign of a coefficient

In the scalar objective $\frac{1}{2}(w-a)^2+\lambda|w|$, the unpenalized estimate is a.

$$\hat{w}=\operatorname{sign}(a)\max(|a|-\lambda,0);\quad a=-3:\ \lambda=1\Rightarrow -2,\ \lambda=4\Rightarrow 0.$$

- For a negative coefficient, shrinkage moves it upward toward zero; it does not subtract the penalty regardless of sign.
- The zero solution is permitted when the evidence magnitude does not exceed the threshold.
- Check the loss normalization before comparing software penalty values. A sparsity result states what this fitted objective selected, not that the omitted variable has no real-world relevance.

Scaling and the intercept change what the penalty means

$$z_j = \frac{x_j - \mu_j}{s_j}, \qquad f(z) = b + \sum_j w_j z_j.$$

- A unit increase now means one training-set standard deviation. Coefficient penalties compare similar scales.
- Fit μ_j, s_j on each training fold and reuse them on its validation fold.
- Penalize slopes w_j; usually leave the offset b unpenalized.
- Changing the response unit rescales fitted coefficients and loss. Reconsider the numerical penalty strength accordingly.

Interpretation

The coefficient for height in millimeters is 1,000 times smaller than the coefficient for height in meters, although the physical prediction is identical.

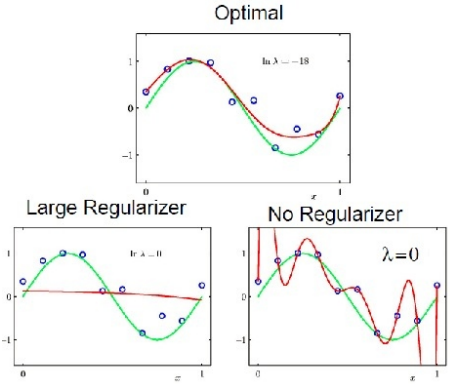
Unit changes can silently change a coefficient penalty

The same physical relation is written as $y = 2x_{\mathrm{m}}$ or $y = 0.02x_{\mathrm{cm}}$, since $x_{\mathrm{cm}} = 100x_{\mathrm{m}}$.

$$\Omega_{\mathrm{m}} = 2^2 = 4, \qquad \Omega_{\mathrm{cm}} = 0.02^2 = 0.0004.$$

- Prediction is identical, yet the numerical squared penalty differs by a factor of 10,000.
- To express the same physical penalty without scaling, the centimeter-coordinate price would need to be 10,000 times larger.
- Standardize using training statistics or define a scale-aware penalty. Equal numerical coefficients and equal penalty prices have no meaning until the measurement units are stated.

The same degree-nine family can produce different fits



- No penalty: coefficients may chase individual noise displacements.
- Moderate penalty: the broad rise and fall remains while unstable oscillation is reduced.
- Very strong penalty: a nearly flat fit can miss the signal.

Compare honestly

Degree stays fixed; only λ changes. Choose it using held-out loss rather than the visual appeal of the training fit.

Choose a stable validation region rather than the training minimum

Illustrative five-fold MSEs, with degree nine fixed:

λ	Training MSE	Mean validation MSE
0	0.000	0.420
10^{-3}	0.018	0.115
10^{-2}	0.041	0.110
10^{-1}	0.120	0.160

The lowest training error favors zero penalty. Validation favors moderate penalties; 0.115 and 0.110 may be indistinguishable given fold variation. Inspect stability, refit the selected setting on development data, then evaluate the fixed procedure on the independent test set.

A one-standard-error choice expresses a conservative selection rule

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Suppose validation mean MSE is 0.10 with estimated standard error 0.02 at the numerical minimum.

$$\text{Acceptable mean-loss threshold} = 0.10 + 0.02 = 0.12.$$

- A stronger penalty with mean loss 0.11 lies within this threshold; a much stronger penalty with 0.16 does not.
- The one-standard-error rule can select the simpler eligible candidate instead of chasing a tiny numerical improvement.
- It is a selection heuristic, not proof of equality or an independent hypothesis test. State how uncertainty was estimated and assess the selected procedure on separate evidence.

A prior gives another interpretation of a penalty

With independent Gaussian observation noise and Gaussian coefficient prior,

$$y_i \mid w \sim \mathcal{N}(f_w(x_i), \sigma^2), \quad w_j \sim \mathcal{N}(0, \tau^2),$$

$$-\log p(w \mid D) = \frac{1}{2\sigma^2} \sum_i (y_i - f_w(x_i))^2 + \frac{1}{2\tau^2} \sum_j w_j^2 + \text{constant}.$$

Maximizing the posterior yields an L2-penalized fit. A smaller prior scale τ favors smaller coefficients. A Laplace prior produces an L1 penalty. This interpretation describes a modeling assumption; it does not establish that the assumed prior is true.

The prior interpretation includes noise scale and sample count

Use Gaussian response noise variance σ^2 and independent Gaussian coefficient prior variance τ^2.

$$\min_w \mathrm{RSS} + \frac{\sigma^2}{\tau^2}\|w\|^2 \iff \min_w \frac{\mathrm{RSS}}{n} + \frac{\sigma^2}{n\tau^2}\|w\|^2.$$

- For $n = 100, \sigma^2 = 0.25, \tau^2 = 1$, the average-loss penalty coefficient is 0.0025.
- A fixed probabilistic prior has a different numerical coefficient under summed and averaged objectives.
- Discarding constants is valid only for terms independent of the parameter currently being optimized. If noise variance is also estimated, revisit which terms depend on it.

Capacity can be limited in several ways

Control	What is restricted
Polynomial degree	Which basis functions may participate
Coefficient penalty	How strongly available features may be used
Tree depth	How many successive distinctions can be made
Early stopping	How far an iterative fitting process follows the data

These choices affect different model families. Early stopping is chosen using validation; a low training objective alone does not specify the best stopping time. None of these controls repairs a split that leaks labels or changes the target population.

Early stopping chooses a point on the fitting trajectory

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An illustrative iterative model produces the following train/validation MSE pairs.

$$t = 1 : (0.40, 0.42), \quad t = 5 : (0.12, 0.18), \quad t = 20 : (0.03, 0.29).$$

- Training loss continues to improve, but validation loss is lowest among these checkpoints at step five.
- Choosing the stopping time uses validation information and is therefore part of model selection.
- Do not confuse a useful stopping decision with an optimizer that has accidentally failed. Compare checkpoints under the same held-out design, then assess the fixed stopping rule independently.

Bias and variance compare models fitted on different samples

Fix one input x. Repeatedly draw a training set D and fit $\hat{f}_D$.

$$f^*(x) = \mathbb{E}[Y \mid x], \qquad \bar{f}(x) = \mathbb{E}_D[\hat{f}_D(x)].$$

- Bias: $\bar{f}(x) - f^*(x)$, systematic displacement of the average fit.
- Variance: $\mathbb{E}_D[(\hat{f}_D(x) - \bar{f}(x))^2]$, sensitivity to which sample was observed.
- Noise: variation in a new response even with its true mean known.

Changing only the random optimizer seed is not the same as repeating the training sample. Here the source of randomness is the data itself.

Zero bias can coexist with large variance

At a fixed input, let the true conditional mean be one and four equally likely trained predictions be $0, 0.5, 1.5, 2$.

$$\bar{f} = 1, \quad \mathrm{Bias}^2 = 0, \quad \mathrm{Var}(\hat{f}) = \frac{1 + 0.25 + 0.25 + 1}{4} = 0.625.$$

- The average predictor is exactly centered, but individual training samples can produce quite different answers.
- With test-noise variance 0.09, expected squared prediction error is 0.715.
- A low bias statement concerns the mean across training experiments. It does not guarantee that the one model fitted today is accurate or stable.

Derive the squared-error decomposition

Let $Y = f^*(x) + \epsilon$, with zero-mean test noise independent of D.

$$\hat{f}_D(x) - Y = \underbrace{\hat{f}_D(x) - \bar{f}(x)}_{A} + \underbrace{\bar{f}(x) - f^*(x)}_{B} - \epsilon.$$

$$\mathbb{E}_{D,\epsilon}[(\hat{f}_D(x) - Y)^2] = \underbrace{\mathbb{E}_D[A^2]}_{\text{variance}} + \underbrace{B^2}_{\text{bias squared}} + \underbrace{\mathbb{E}[\epsilon^2]}_{\text{noise}}.$$

- $\mathbb{E}_D[A] = 0$, so $\mathbb{E}[AB] = 0$.
- Noise has mean zero and is independent, so its cross terms vanish.
- This is an expected squared-error identity at a fixed input, not a formula for one realized residual.

Independent test noise is essential to the usual decomposition

A model memorizes one training response $y = f^*(x) + \epsilon$ at its observed input.

$$\hat{f}(x) = y \Rightarrow (\hat{f}(x) - y)^2 = 0; \qquad Y_{\text{new}} = f^*(x) + \epsilon_{\text{new}}.$$

- The zero training residual reuses the same noise in fit and evaluation; it is not an independent prediction error.
- With independent zero-mean noises of variance σ^2, the new squared error has expectation $2\sigma^2$.
- The model has carried training noise into its prediction while the new response adds fresh noise. This is why cross terms must be justified rather than silently dropped.

A numerical decomposition makes the three terms tangible

At a fixed input, suppose $f^*(x) = 1$, test-noise variance is 0.09, and four equally likely fitted predictions are $0.6, 0.8, 1.0, 1.2$.

$$\bar{f} = 0.9, \quad \text{Bias}^2 = (0.9 - 1)^2 = 0.01,$$

$$\text{Variance} = \frac{(-0.3)^2 + (-0.1)^2 + (0.1)^2 + (0.3)^2}{4} = 0.05.$$

$$\mathbb{E}[(\hat{f} - Y)^2] = 0.01 + 0.05 + 0.09 = 0.15.$$

Even a perfectly centered, stable predictor cannot remove the 0.09 observation noise under this model. The four predictions are an illustrative finite distribution of training outcomes.

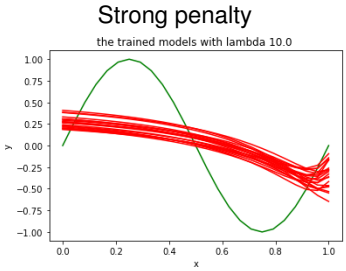
Compare total error rather than optimizing only one component

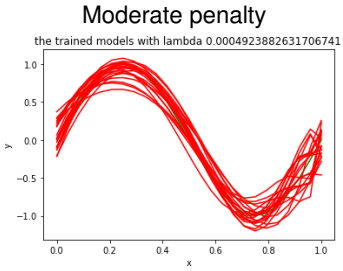
At the same input, all three candidates face observation-noise variance 0.09.

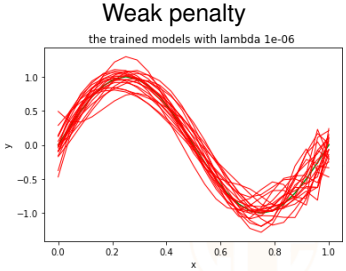
$$A : 0.16 + 0.01 + 0.09 = 0.26; \quad B : 0.0025 + 0.18 + 0.09 = 0.2725; \quad C : 0.01 + 0.04 + 0.09 = 0.14.$$

- The first two terms are bias squared and variance. A is stable but systematically displaced; B is well centered but unstable.
- C has neither the smallest bias nor the smallest variance, yet it has the lowest total expected squared error.
- These quantities are available in a controlled simulation. Real tasks generally require indirect diagnosis from held-out behavior rather than directly observed oracle components.

Repeated fits reveal stability and systematic mismatch







- Closely clustered curves can still miss the true signal: low variance, high bias.
- Wide variation across fitted curves indicates sensitivity to the observed sample.
- Compare their mean location and spread separately. One attractive fit cannot establish the behavior across repeated samples.

Averaging helps most when fitted errors are not identical

For K equally weighted predictors with common prediction variance v and pairwise correlation ρ, consider their average.

$$\mathrm{Var}(\bar{f}_K) = v\left[\rho + \frac{1-\rho}{K}\right].$$

- With $v = 0.16, K = 4, \rho = 0.5$, the average variance is 0.10, below 0.16.
- Independent fluctuations would give $v/K = 0.04$; identical fluctuations would give no reduction.
- Averaging does not automatically remove a shared systematic bias. Later ensemble lectures exploit this connection through deliberately varied fitted models.

The bias–variance tradeoff is a guide, not an automatic law

$$\text{Expected squared error} = \text{bias}^2 + \text{variance} + \text{noise}.$$

- More flexibility often reduces approximation bias but increases sensitivity on a fixed small sample.
- Stronger regularization often increases bias and lowers variance.
- More representative data can lower estimation variability without forcing a simpler model.
- These are useful tendencies, not universal monotonic laws for every algorithm or regime.

Practical limit

The true conditional mean and all possible training samples are unavailable. Diagnose using observable training/validation behavior; do not label a single numerical gap as an exact variance estimate.

The noise floor is conditional on the information supplied

The irreducible variance is $\mathrm{Var}(Y \mid X)$, relative to the available input description.

$$\mathrm{Var}(Y \mid X) \text{ can differ from } \mathrm{Var}(Y \mid X, Z).$$

- A useful additional measurement Z may explain variation that appeared unpredictable using X alone.
- Improved measurement precision can also reduce observation noise; more fitting iterations cannot repair an unreliable sensor.
- Do not promise that a richer input always helps: it must be informative, available at prediction time, and evaluated without leakage. Noise is not a fixed excuse for every remaining error.

Learning curves vary sample size, not model degree

Fit the same model and hyperparameters on increasing training subsets; evaluate each fitted model on held-out observations.

Training size	Training MSE	Validation MSE	Gap
20	0.02	0.40	0.38
80	0.06	0.22	0.16
320	0.09	0.13	0.04

The shrinking gap suggests that additional data helps this estimator. Curves need not be perfectly monotone in one finite experiment. Repeat splits where appropriate and show variability. Do not confuse this sample-size axis with a validation curve indexed by λ or polynomial degree.

Two high, close curves suggest a different remedy from a large gap

The same low-flexibility model is fitted using increasingly many representative observations.

$$n = 20 : (0.36, 0.43), \quad n = 80 : (0.38, 0.42), \quad n = 320 : (0.40, 0.41).$$

- Pairs show training and validation MSE. The gap closes, but the common error level remains high.
- This suggests that adding similar data alone may have limited benefit; inspect the model family, features, labels, and optimization.
- Compare a better representation under the same evaluation design. High training error is a diagnostic clue, not a proof that the true conditional function is complicated.

Use the shape of the curves to choose the next intervention

Observed behavior	Plausible diagnosis	Next comparison
Both errors high and close	Family misses useful structure; or labels/features are poor	Better features, more suitable family, verify data
Training low; validation high	Sample-sensitive fit; or split mismatch	Stronger regularization, more representative data, inspect split
Both errors low but future fails	Evaluation target differs from future use	Time/group-aware evaluation, examine distribution shift

These are hypotheses to test, not diagnoses established by a single graph. First confirm optimization is adequate and the held-out observations were truly excluded.

Paired fold comparisons remove one source of irrelevant variation

Compare two candidates on the same four validation folds, rather than assigning different splits to them.

$$R_A = (0.20, 0.30, 0.10, 0.40), \quad R_B = (0.18, 0.28, 0.11, 0.37), \quad \overline{R_A - R_B} = 0.015.$$

- B improves three folds and worsens one. The mean difference is an improvement of 0.015, not improvement on every fold.
- Matching folds makes each difference reflect the same held-out observations, reducing accidental split difficulty differences.
- Folds share training data, so these differences are not automatically independent trial results. Inspect their size and variability before declaring a decisive gain.

Period 2 exercise: distinguish bias from instability

Candidate	Training MSE	Validation MSE	Change after more data
A: degree 1	0.42	0.44	Little change
B: degree 9, no penalty	0.00	0.39	Validation falls to 0.15
C: degree 9, moderate penalty	0.06	0.12	Stable around 0.11

1. Explain why increasing the iteration count is not the first remedy for A if its objective has converged.
2. Explain why B benefits from more data even though its training fit becomes worse.
3. Select a candidate and name the independent evidence still needed.

Solution: design an experiment that distinguishes two explanations

A high-validation-error model might lack useful structure or might be too sensitive to a small sample.

- First verify convergence and leakage-free evaluation. Then compare a more expressive family and a more strongly regularized version on the same folds.
- If added representative data mainly reduces a large train/validation gap, sample sensitivity is supported; if both remain high, inspect representation and labels.
- A useful diagnosis predicts what a controlled change should improve. Changing family, data, and splitting simultaneously obscures the explanation.

A fixed rule has a concentration guarantee

For a classifier fixed independently of an i.i.d. sample, let $Z_i = \mathbf{1}\{f(x_i) \neq y_i\} \in [0,1]$.

$$\hat{R}(f) = \frac{1}{n}\sum_i Z_i, \qquad R(f) = \mathbb{E}[Z_i].$$

Hoeffding's inequality gives

$$P\big(|\hat{R}(f) - R(f)| \geq \varepsilon\big) \leq 2e^{-2n\varepsilon^2}.$$

- Larger samples make a large deviation less likely.
- This statement requires bounded loss and independent sampling.
- It applies directly to a model frozen before an independent test set; a training-selected rule needs further reasoning.

The range of the loss changes a concentration bound

Hoeffding applies to losses bounded in a declared interval $[a, b]$.

$$P(|\hat{R} - R| \geq \varepsilon) \leq 2 \exp\left[-\frac{2n\varepsilon^2}{(b-a)^2}\right].$$

- Zero–one loss uses range one. A loss in $[0, 5]$ has a larger tolerance at the same sample count and confidence.
- Squaring an unrestricted real residual is not bounded; extra assumptions or another concentration argument are needed.
- Clipping squared loss can restore a bound but changes the risk being estimated. State the clipped objective rather than pretending it is the original unlimited loss.

Convert a probability bound into an error tolerance

Set $2e^{-2n\varepsilon^2} = \delta$ and solve:

$$\varepsilon = \sqrt{\frac{\log(2/\delta)}{2n}}.$$

Independent test

With $n = 200$ and $\delta = 0.05$, $\varepsilon \approx 0.096$. If test error is 0.10, then with probability at least 0.95, $R(f) \leq 0.196$.

- Confidence concerns repetition of the sampling experiment, not a guarantee for every dataset.
- The bound can be conservative; it does not assert that the true error equals its upper endpoint.
- The evaluation must still represent the intended future population.

Confidence and tolerated deviation are separate choices

For one fixed classifier, compare the Hoeffding tolerance at different counts and confidence levels.

$$\varepsilon = \sqrt{\frac{\log(2/\delta)}{2n}}: \quad (n, \delta) = (200, .05) \Rightarrow .096,\ (800, .05) \Rightarrow .048,\ (200, .01) \Rightarrow .115.$$

- Four times the count halves the tolerance when confidence is fixed.
- Requesting 99% confidence rather than 95% produces a wider tolerance with the same data.
- The interval is a repeated-sampling guarantee under the assumptions. It does not convert confidence into the probability that an individual future label is correct.

Choosing a rule after observing the sample changes the problem

Imagine checking M predeclared rules on the same sample and choosing the smallest empirical error.

- Each rule can experience a sampling fluctuation.
- The selected winner may be the rule with the most favorable fluctuation.
- A guarantee for one pre-fixed rule does not automatically cover a rule selected from this sample.

Required event

Control the deviation for every candidate simultaneously: $\sup_{f \in \mathcal{H}} |\hat{R}(f) - R(f)| \leq \varepsilon$.
If all candidates are covered, the selected candidate is covered too. This is the reason capacity enters learning theory, rather than only the observed training score.

Many opportunities for a lucky result increase selection risk

As an illustration, suppose 20 independent candidate checks each have a 5% probability of a misleading favorable event.

$$P(\text{at least one}) = 1 - (1 - .05)^{20} \approx 0.642.$$

- The independence assumption belongs to this illustrative calculation; real candidate scores often depend on the same observations.
- Without independence, a union bound still limits the probability by the sum of event probabilities, capped at one.
- A small risk per check does not remain small when many results are inspected. Model selection therefore requires joint control or genuinely independent final assessment.

A finite class pays a logarithmic selection cost

For M predeclared classifiers, union-bound the bad events:

$$P\left(\exists f:\ |\hat{R}(f) - R(f)| \geq \varepsilon\right) \leq \sum_{f=1}^{M} 2e^{-2n\varepsilon^2} = 2Me^{-2n\varepsilon^2}.$$

Therefore, with probability at least $1 - \delta$, simultaneously

$$R(f) \leq \hat{R}(f) + \sqrt{\frac{\log(2M/\delta)}{2n}}.$$

For $M = 100, n = 1000, \delta = 0.05$, the tolerance is about 0.0644; a training error of 0.03 has upper bound 0.0944. All candidates must be specified independently of the sample used for this guarantee.

Uniform control turns ERM into an excess-risk statement

Suppose every candidate satisfies $|\hat{R}(f) - R(f)| \le \varepsilon$, and $\hat{f}$ minimizes empirical risk in the class.

$$R(\hat{f}) \le \hat{R}(\hat{f}) + \varepsilon \le \hat{R}(f^*_{\mathcal{H}}) + \varepsilon \le R(f^*_{\mathcal{H}}) + 2\varepsilon.$$

- The first and last steps use uniform deviation control; the middle step uses empirical minimization.
- If optimization is within ζ of the empirical minimum, the final excess-risk bound becomes $2\varepsilon + \zeta$.
- For $\varepsilon = 0.05, \zeta = 0.01$, this gives 0.11 above the familyś population optimum. Approximation error relative to all possible rules remains a separate term.

Sample complexity depends on tolerance and confidence

For the same finite-class guarantee, requiring deviation at most ε gives

$$n \geq \frac{\log(2M/\delta)}{2\varepsilon^2}.$$

Calculation

$M = 100, \delta = 0.05, \varepsilon = 0.05$ requires at least $\lceil 8.294/0.005 \rceil = 1659$ samples under this bound.

- Halving the tolerated gap multiplies this sufficient sample size by four.
- Increasing confidence or candidate count increases the requirement logarithmically.
- This is a sufficient bound under stated assumptions, not an exact minimum or a universal data collection prescription.

Increasing candidate count has a measurable but logarithmic cost

Hold $n = 1000$ and $\delta = 0.05$ fixed in the finite-class deviation tolerance.

$$M = 100 : \ \varepsilon \approx 0.0644; \qquad M = 10000 : \ \varepsilon = \sqrt{\log(400000)/2000} \approx 0.0803.$$

- A hundredfold increase in candidate count raises the sufficient tolerance, although it does not multiply it by one hundred.
- More selection freedom requires a stronger joint statement. Candidates must be specified independently of the sample used in the bound.
- This comparison does not justify unlimited adaptive searching. New data-driven features can enlarge the effective procedure beyond the stated finite list.

Infinite parameter values can produce few label patterns

Consider one-dimensional thresholds $f_t(x) = \mathbf{1}\{x \geq t\}$, where t may be any real number.

- There are infinitely many parameter values.
- On three ordered points $x_1 < x_2 < x_3$, only four patterns occur: $111, 011, 001, 000$.
- Moving t between the same adjacent points changes no predicted label.

Capacity idea

Count distinguishable labelings on a finite set, rather than all numerical parameter choices.

The pattern 101 cannot occur for a positive-ray threshold. To quantify this limitation, we next define shattering and VC dimension.

Threshold growth is finite even with infinitely many cut points

On n distinct ordered points, a positive-ray threshold can place its transition in $n+1$ gaps.

$$\Pi_{\text{threshold}}(n) = n+1; \qquad n = 5: \ 6 \text{ patterns versus } 2^5 = 32 \text{ possible labelings.}$$

- Each realized pattern has one contiguous suffix of positives, including the empty and full suffixes.
- Most labelings are unavailable; 10101 on five ordered points cannot be represented.
- Capacity counts distinguishable label patterns, not the infinitely many decimal threshold values. This explicit growth calculation explains why the infinite numerical parameter does not force an infinite selection penalty.

Shattering requires every labeling of the same point set

A class shatters n points if it can realize all 2^n binary labelings of those points.

Positive-ray thresholds	Can the class realize every labeling?
One point	Yes: move threshold to either side, producing 0 or 1
Two ordered points	No: 00, 01, and 11 occur; 10 does not

VC dimension

The largest size of a point set that can be shattered. Positive-ray thresholds have VC dimension 1.

Getting one particular labeling correct is not shattering. The class must accommodate every possible labeling of that same set.

Allowing either orientation changes one-dimensional capacity

Now allow affine one-dimensional classifiers $\mathbf{1}\{wx + b \geq 0\}$ with either sign of w.

- Two ordered points can realize 00, 01, 10, and 11: either threshold orientation or a constant label suffices.
- Three ordered points cannot realize 010 or 101 with a single threshold transition.
- This class has VC dimension 2, unlike fixed positive-ray thresholds with VC dimension 1. State exactly which hypothesis class is being counted before quoting its dimension.

Intervals have more capacity than positive-ray thresholds

Let $f_{a,b}(x) = \mathbf{1}\{a \le x \le b\}$.

- Two points can realize 00, 01, 10, and 11 by choosing an interval appropriately.
- For three ordered points, labeling 101 is impossible: an interval containing both endpoints also contains the middle point.
- Thus the interval class has VC dimension 2.

Comparison

The input dimension remains one. The permitted shape of the positive region changed, so capacity changed.

VC dimension is a property of a hypothesis class. Counting input features or coefficients alone does not always reveal it.

Two disjoint positive intervals can shatter four ordered points

Consider the class whose positive region is the union of at most two intervals on the real line.

- Every binary labeling of four ordered points has at most two runs of positive points; two intervals can cover those runs.
- Five ordered points labeled 10101 have three separated positive runs. Two intervals cannot include them while excluding the two intervening negatives.
- Thus the VC dimension is 4. More permitted positive components increase expressive capacity, although the input dimension remains one.

Affine lines in the plane shatter three noncollinear points

For $f_{w,b}(x) = \mathbf{1}\{w^\top x + b \geq 0\}$ in two dimensions:

- Three noncollinear points form a triangle. Each one-versus-two labeling can be separated by a line; constant labels also occur.
- Four convex-position points cannot realize alternating labels around the quadrilateral: the two positive and two negative convex hulls intersect.
- If one point lies inside a triangle, labeling the center opposite to all three vertices also fails.

Hence affine lines have VC dimension 3. In d dimensions, unrestricted affine hyperplanes have VC dimension $d + 1$. Including the intercept matters.

The XOR labeling makes the four-point limitation explicit

In the plane, label $(0,0),(1,1)$ positive and $(0,1),(1,0)$ negative.

$$b \geq 0, \quad w_1 + w_2 + b \geq 0; \qquad w_2 + b < 0, \quad w_1 + b < 0.$$

- Adding the positive inequalities requires $w_1 + w_2 + 2b \geq 0$; adding the negative inequalities requires it to be negative.
- The requirements contradict each other, so one affine line cannot realize this labeling.
- This is a limitation of the linear hypothesis class, not a failure to find its parameters. Feature expansion or a richer class changes the representable patterns and must be evaluated accordingly.

VC theory replaces an infinite count with a growth function

Let $\Pi_{\mathcal{H}}(n)$ be the maximum number of labelings realizable on n points. For VC dimension $d < n$, the Sauer bound gives

$$\Pi_{\mathcal{H}}(n) \leq \sum_{j=0}^{d} \binom{n}{j} \leq (en/d)^d.$$

A typical uniform classification bound has the order

$$R(f) \lesssim \hat{R}(f) + O\left(\sqrt{\frac{d \log(en/d) + \log(1/\delta)}{n}}\right).$$

- More data reduces the capacity term; a richer class increases it.
- Constants and additional conditions belong to a specific theorem. This order expression is not a measured test-error equation.

A growth-function bound can be checked with small integers

For VC dimension $d = 2$ and ten points, compare the unrestricted label count with the Sauer bound.

$$\sum_{j=0}^{2} \binom{10}{j} = 1 + 10 + 45 = 56; \qquad 2^{10} = 1024, \quad (10e/2)^2 \approx 184.7.$$

- The combinatorial bound permits at most 56 patterns; the simpler power bound is looser but easier to manipulate.
- An upper bound need not equal the class's exact growth function. Different classes with the same VC dimension can have different realized counts.
- The main consequence is controlled growth rather than every possible labeling. This control is then inserted into a uniform risk argument with its stated sampling conditions.

Structural risk minimization compares fit and capacity together

Consider nested classes $\mathcal{H}_1 \subset \mathcal{H}_2 \subset \mathcal{H}_3$.

Class	Empirical risk	Capacity penalty	Sum
$\mathcal{H}_1$	0.18	0.04	0.22
$\mathcal{H}_2$	0.07	0.09	0.16
$\mathcal{H}_3$	0.01	0.20	0.21

The illustrative criterion selects $\mathcal{H}_2$. Added flexibility improves fit, but its uncertainty cost can exceed that gain. Theory motivates this comparison; in practice validation often selects degree or penalty because distribution-free bounds are conservative. A penalty criterion is not the same thing as observed test loss.

A larger sample can justify a richer selected family

Illustrative empirical risks are $0.10, 0.05, 0.02$ for three nested families. Compare capacity terms as count increases.

$$n = 100 : \ (0.05, 0.10, 0.20); \quad n = 400 : \ (0.025, 0.05, 0.10); \quad n = 1600 : \ (0.0125, 0.025, 0.05).$$

- Fit plus capacity at $n = 400$ is $(0.125, 0.10, 0.12)$, selecting the middle family; at $n = 1600$ it is $(0.1125, 0.075, 0.07)$, selecting the richest.
- Only the illustrative capacity term was changed here to isolate the mechanism; actual empirical fits also change with data.
- Complexity is judged relative to evidence, not by a permanent rule that smaller models are always preferable.

Simplicity is useful when its assumptions fit the problem

- No-free-lunch results average over very broad problem families. They do not prevent useful comparisons on a specified real task.
- Occam's principle favors unnecessary complexity being removed while retaining adequate fit.
- Minimum description length balances the description of the model with the remaining unexplained data: $L(\text{model}) + L(\text{data} \mid \text{model})$.
- A simple wrong rule can require a long description of its residual errors; a more informative representation may shorten the total.

Boundary

These principles guide model design. They do not replace independent evaluation or guarantee robustness to a changed population.

Description length balances model size and unexplained data

Compare two coding schemes for the same observations: the code for the model plus the code for its remaining errors.

$$L_A = 20 + 5 = 25, \qquad L_B = 5 + 50 = 55 \quad \text{bits in an illustrative code.}$$

- A has a longer model description but explains the data well; B is simpler yet leaves a long residual description.
- Minimum description length favors the smaller total, not merely the shortest model.
- A coding scheme must be specified to interpret its lengths. This principle motivates useful complexity control but does not substitute for held-out prediction assessment or establish a particular real-world error rate.

Synthesis: explain a result and choose a justified next step

Case

A degree-nine model fits 10 noisy observations exactly. Validation MSE is high. With 100 representative observations it improves; moderate L2 regularization also improves it.

1. Explain the likely role of sample sensitivity without claiming a directly measured variance.
2. Specify an honest procedure for choosing degree and λ.
3. State the independence, population, and bounded-loss conditions behind a classification generalization guarantee.

Bridge to linear models

Next lecture derives how least squares and logistic likelihood actually choose parameters. The same validation and regularization principles apply to both.

Take-home: fit, selection, assessment, and theoretical assumptions answer different parts of the learning problem.

Solution: separate the evidence from the claim

A degree-nine fit improves with more data and moderate regularization. Explain what is supported and what remains to be tested.

- The changes support a sample-sensitivity explanation; they do not directly measure a population bias–variance decomposition from one dataset.
- Select degree and penalty inside development folds; test a frozen pipeline on an independent sample from the intended population.
- If future inputs shift to a new sensor, create a matching evaluation design. An i.i.d. classification bound for the old population cannot guarantee performance on the new one.